

TRAN VAN MANH

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EDUCATION

Master of Science in Control and Robot Engineering

Chungbuk National University

Cheongju, South Korea

March 2022 - August 2024

- GPA: 3.69/4.0
- Thesis: Autonomous Navigation in Complex Environments Based on Deep Reinforcement Learning Using Selective Policy Switching for Mobile Robots.

Bachelor of Science: Control Engineering and Automation

Hanoi University of Science and Technology

Hanoi, Vietnam

August 2015 - August 2020

- Ranked 1st in Vietnam for Electrical and Electronic Engineering by QS World University Rankings.
- GPA: 3.12/4.0

EXPERIENCE

Robotics Engineer

ROBOT GLOBAL TEAM (RGT) company

Seoul, South Korea

September 2024 – Present

- Developing local planners for autonomous navigation using sampling and search-based methods.
 - Developing motion planning methods on service robot, multi-robot system, improving computation cost on SBC by 30%, ensuring smooth driving and control.
 - Releasing software versions at restaurants.
- Developed and maintained the motor controller for service robots.
 - Developing parts of motor controllers, such as an emergency button and stable serial port connection.
 - Maintain and handle the safety issues of the motor driver, log system file, voltage percentages, and filter out the voltage of the battery.

Master Student/ Researcher at Intelligent Robot Laboratory

Chungbuk National University

Cheongju, South Korea

March 2022 – August 2024

ATC-project: Development and implementation of a navigation algorithm with collision avoidance based on deep reinforcement learning (Deep-RL).

- Successfully transferred deep reinforcement learning models from simulation training to the real world for service human robots.
- Evaluated mapless navigation and localization with a motion capture system, achieved over 95% navigation success rate and localization error less than 10%, certified by KR.

Researcher

Vietnam-Korea Institute of Science and Technology

Hanoi, Vietnam

August 2020 – February 2022

- Research and development of autonomous mobile robot technology for industrial transport applications.
 - Defined functions, requirements, specifications, and technical criteria for the prototype of autonomous robot system.
 - Developed and designed kinematic model and autonomous navigation framework for omnidirectional mobile robot prototype.
- Designed and optimized magnetic flux density in PMSM motor on JMAG.
- **Frameworks used:** Optimization using genetic algorithm, navigation-stack framework, AMCL, Kalman filter.

Research Intern
Vietnam Academy of Science and Technology

Hanoi, Vietnam
March 2018 – August 2020

Participating in Project: 'Smart Human-Form IVASTBot Robot Applied in Communication and Serving Humans', phase 01.

- Designed an intelligent trajectory system for an omnidirectional mobile robot.
- Designed backstepping control, sliding mode control algorithms in MATLAB.
- Deploy low-level control in C for embedded systems on omnidirectional mobile robots.

SKILLS

- **Programming language:** C++, C, Python, MATLAB.
- **Frameworks:** ROS/ROS2, OpenCV, Rtab-map.
- **Simulation tools and Libraries:** Gazebo, Stage, Pytorch, NumPy, Matplotlib, gym, LaTeX, GitHub.
- **Soft skills:** Problem solving, teamwork, leadership.

PUBLICATIONS AND PATENTS

Publications

- 2024/07, "Cooperative Deep Reinforcement Learning Policies for Autonomous Navigation in Complex Environments," in *IEEE Access*, vol. 12, pp. 101053-101065, 2024, doi: 10.1109/ACCESS.2024.3429230. (**First author**)
- 2024/03, "Mapless Navigation with Distributional Reinforcement Learning", Journal of Korea Robotics Society, Korea. (**First author**)
- 2021/04, "Adaptive Control for Uncertain Model of Omni-directional Mobile Robot Based on Radial Basis Function Neural Network", *International Journal of Control, Automation, and Systems*. (**Co-author**)
- 2020/04, "Autonomous Navigation for omnidirectional Robot Based on Deep Reinforcement Learning", *International Journal of Mechanical Engineering and Robotics Research*. (**Co-author**)
- 2019/12, "Nonlinear Backstepping-Sliding Mode Control of Electro-Hydraulic Systems", *ICERA 2019 International Conference on Engineering Research and Applications*. (**Co-author**)
- 2019/12, "Mapping and Navigation with Four-Wheeled Omnidirectional Mobile Robot based on Robot Operating System", *2019 International Conference on Mechatronics, Robotics and Systems Engineering (MoRSE)*. (**Co-author**)
- 2019/12, "Neural network based adaptive control of web transport system", *2019 International Conference on System Science and Engineering (ICSSE)*. (**Co-author**)
- 2019/12, "Trajectory tracking control for four-wheeled omni-directional mobile robot using backstepping technique aggregated with sliding mode control", *2019 First International Symposium on Instrumentation, Control, Artificial Intelligence, and Robotics (ICASYMP)*. (**Co-author**)

Patents

- **Van Manh Tran** and Gon-Woo Kim, "Autonomous navigation system in complex environments based on deep reinforcement learning using selective policy switching for a mobile robot," *KR Patent MP24-0134/MP24-0141*, 2024.

ACHIEVEMENT

- Receive a scholarship for four semesters of the master's program at Chungbuk National University, 2022-2024.
- Best paper in the Journal of Korea Robotics Society, Korea, 2024.
- Best graduation thesis defense for undergraduates, Hanoi University of Science and Technology, Vietnam, 2020.